DC Electronic Load Rev D: one board, isolated, USB-C powered

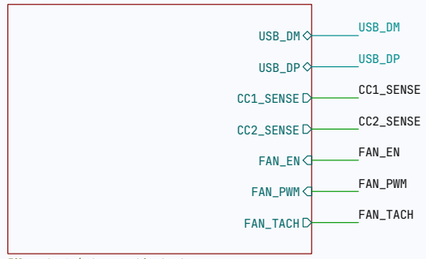
150 W, 0.6 - 58 V, 0 - 10.2 A. Hardware CC and CV, firmware CR / CP / battery. Local 7-segment UI, USB CDC / DFU, SCPI over USB.

Reading order: USB & Power (2), MCU (3), Front Panel (4), Isolated Interface (5), Setpoint & CV (6), Channels (10-13), Sense (7), Protection (8), Power Path (9). Nets here join by name.

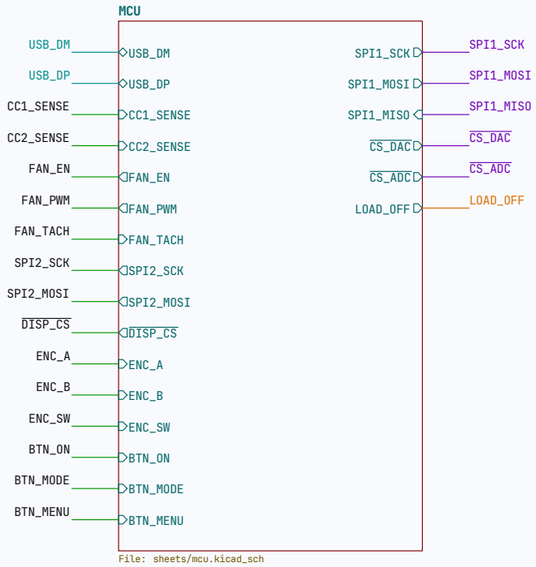
Global power symbols: VBUS, +3V3, +12V_FAN, DGND on the USB side; +12V_ISO, +5V_ISO, AGND, PGND on the analog side. The analog stage is the LTspice-verified Rev C design (../simulation).

USB SIDE (DGND, earth-referenced through the PC)

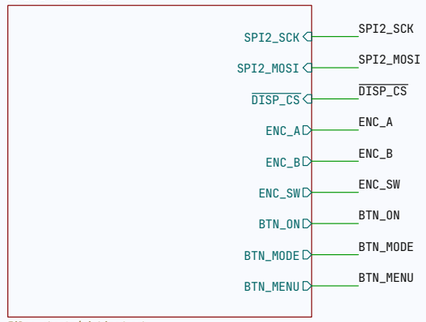
USB & Power



MCU

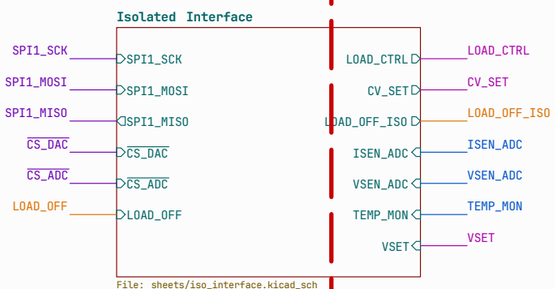


Front Panel UI



ISOLATION

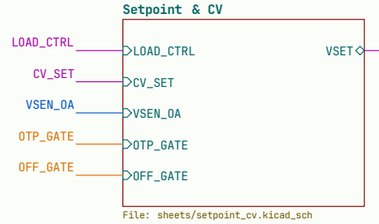
Isolated Interface



ISOLATION BARRIER, 4 mm

ISOLATED ANALOG SIDE (AGND = DUT minus, floats with the DUT)

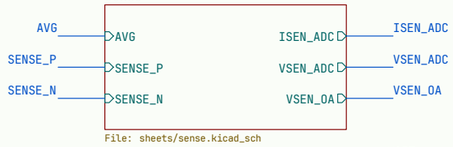
Setpoint & CV



Channel 1



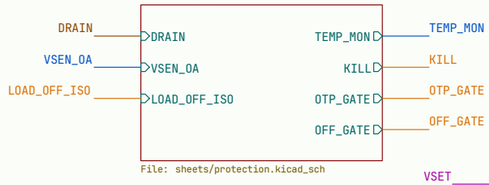
Sense



Channel 2



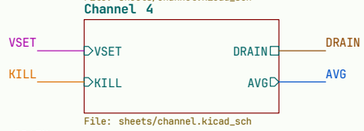
Protection



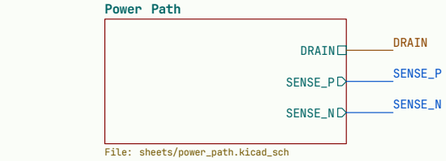
Channel 3



Channel 4



Power Path



NET COLOUR KEY

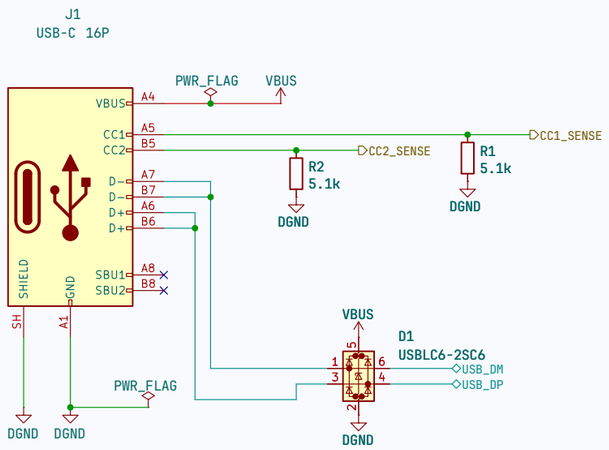
- DUT current path
- set-point (LOAD_CTRL, V_SET, CV_SET)
- measurement (AVG, ISEN, VSEN, V_NTC)
- protection (KILL, OTP, OFF, LOAD_OFF)
- isolated SPI
- USB data
- supply rails
- isolation barrier

DC Load Box	
Sheet: /	
File: DC_Load_RevD.kicad_sch	
Title: DC Electronic Load Rev D: one board, isolated, USB-C powered	
Size: A3	Date: 2026-10-31
KiCad E.D.A. 10.0.6	Id: 1/13

USB-C input and power tree

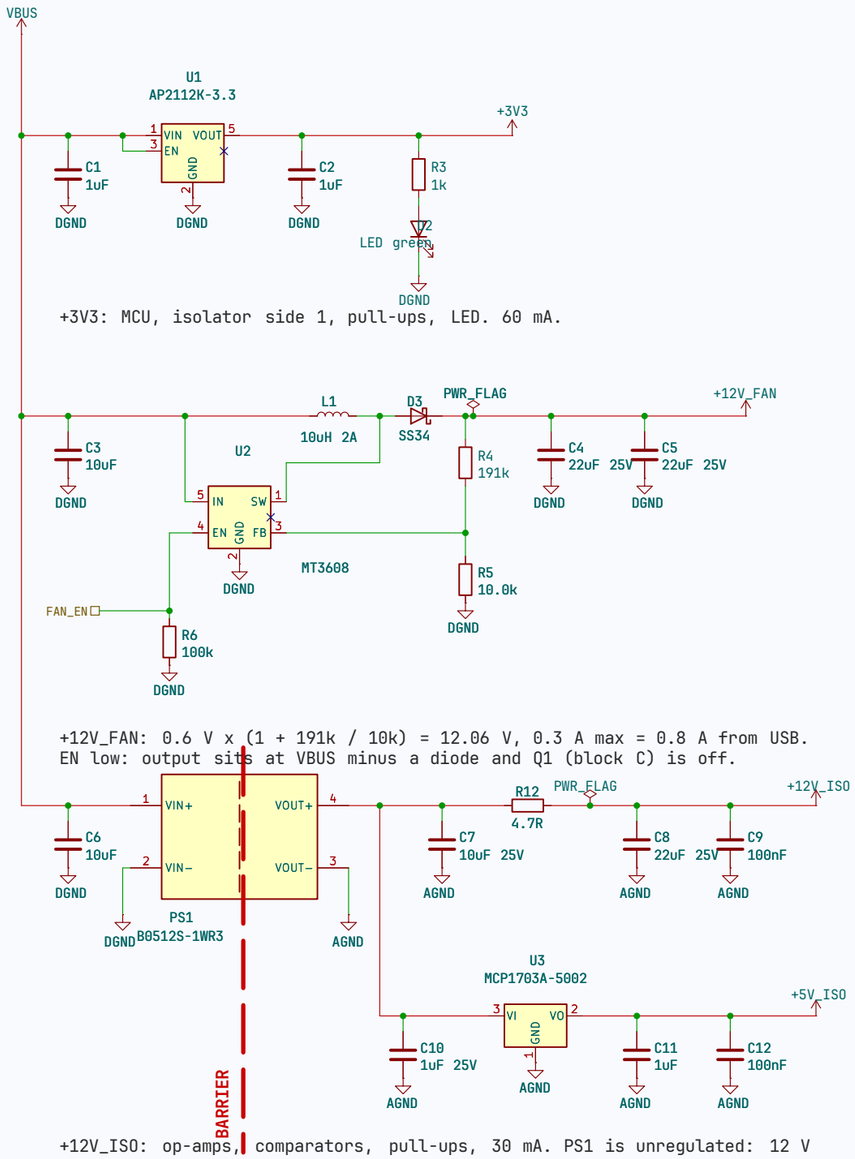
One USB-C port feeds four rails. Left of the barrier is USB ground (DGND); the analog stage floats on AGND = DUT minus.

A USB-C input: 5 V, data, current advertisement (CC)

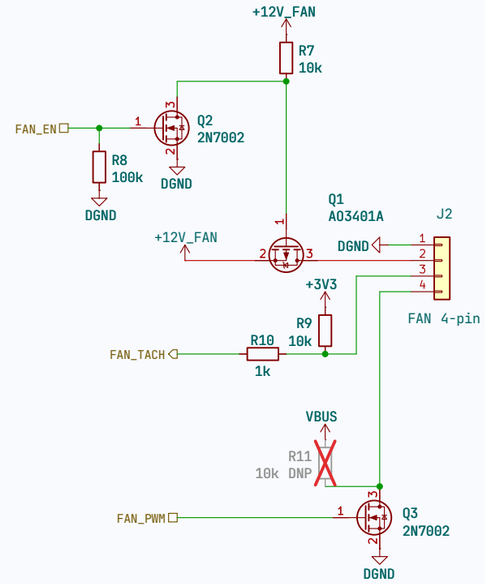


CC1 / CC2 5.1k pull-downs = USB-C sink. The MCU reads the CC voltage to learn the source limit: 0.4 V = 500 mA (legacy source), 0.9 V = 1.5 A, 1.7 V = 3 A. Below 1.5 A firmware keeps the fan slow and limits power to 55 W. Shield to DGND: a metal enclosure is on the USB side; DUT terminals, heatsink hardware and the NTC lug must not touch it.

B Power tree: VBUS 5 V trunk, three branches, four rails

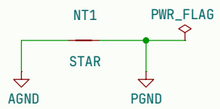


C Fan switch and connector



J2 = PC fan pinout: 1 GND, 2 +12 V, 3 tach (open collector, pulled to 3.3 V), 4 PWM 25 kHz. 2- and 3-wire fans use pins 1 to 3, on/off only. Q3 drives PWM open-drain: firmware inverts the duty. R11 only for fans without a PWM pull-up.

D Ground star point and the isolation barrier rating



AGND (analog reference) meets PGND (DUT minus, source resistors) ONLY at NT1. Place NT1 at the centre of the four Rs bottom pads: 1 mR between an Rs and NT1 is 10 mV at 10 A (4 % of full scale).

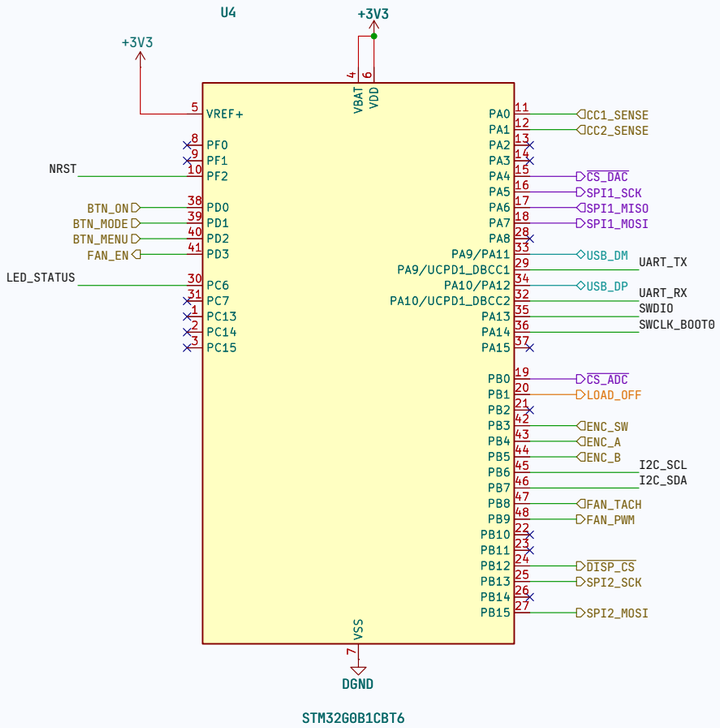
Barrier: DGND side to AGND / PGND side. Rated 150 V DC working (IEC 60664-1 functional, PD2, material group IIIb: 1.6 mm creepage; IPC-2221B B2: 0.7 mm). Board rule 4 mm, no copper crosses on any layer. ISO7761 (5 kVrms, 8 mm) and the 3 kVDC DC-DC exceed it by far. Float the load on a DUT referenced within 150 V of earth; not for mains work.

DC Load Box	
Sheet: /USB & Power/	
File: usb_power.kicad_sch	
Title: USB-C input and power tree	
Size: A3	Date: 2026-10-31
KiCad E.D.A. 10.0.6	Id: 2/13
	Rev: D

MCU: STM32G0B1CBT6 (LQFP48), USB device, SPI to the isolated stage, SPI to the display

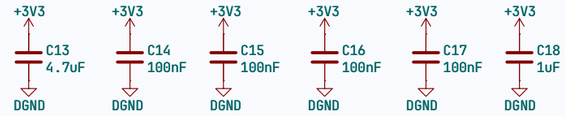
Crystal-less: USB SOF trims the 48 MHz HSI. DFU over USB is in ROM (hold BOOT, tap RESET). Everything here is on DGND.

A Microcontroller



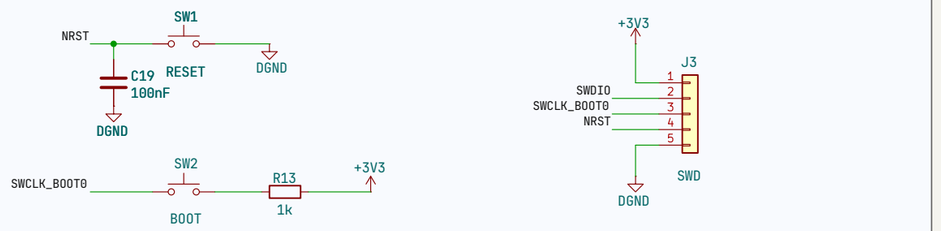
Pin plan (STM32G0B1CBT6, LQFP48)
SPI1 PA5 SCK, PA6 MISO, PA7 MOSI, PA4 /CS DAC, PB0 /CS ADC, PB1 LOAD_OFF (isolator)
SPI2 PB13 SCK, PB15 MOSI, PB12 /CS = MAX7219 LOAD (display chain, via 74AHCT125)
TIM3 PB4 / PB5 encoder, PB3 push. TIM17 PB9 fan PWM, TIM16 PB8 tach, PD3 FAN_EN
ADC PA0 / PA1 = CC1 / CC2. USB PA11 DM, PA12 DP, crystal-less (CRS)
USART1 PA9 / PA10 debug. I2C1 PB6 / PB7 OLED. PD0 ON/OFF, PD1 MODE, PD2 MENU, PC6 LED
Why not the STM32F103RBT6 of Rev B: no USB DFU in its ROM (USART only), needs a crystal, no CRS, same LQFP64 size. The G0B1 is cheaper, smaller, and its ROM DFU removes the second USB port.
Spare: PA2, PA3, PA8, PA15, PB2, PB10, PB11, PB14, PC7, PC13-PC15, PF0, PF1 (NC).

B Decoupling



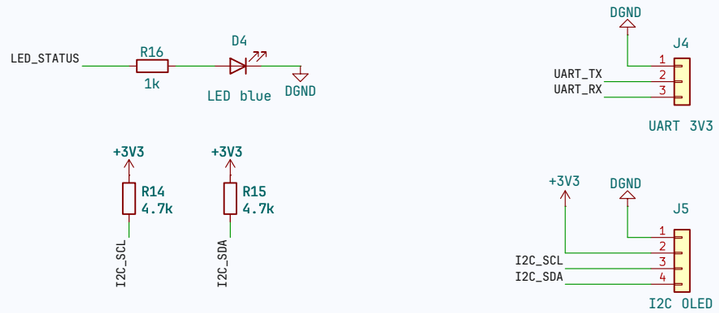
All X7R. One 100 nF per VDD pin, 1 uF + 100 nF at VREF+ (C17, C18).
Place within 2 mm of the pins.

C Reset, BOOT0 (DFU), SWD



BOOT0 shares PA14 (SWCLK) on the STM32G0. Hold BOOT, tap RESET: the ROM bootloader enumerates as USB DFU. Once, with STM32CubeProgrammer, clear option bit nBOOT_SEL so the pin is honoured (the factory default ignores it).

D Debug UART, I2C expansion (OLED), status LED



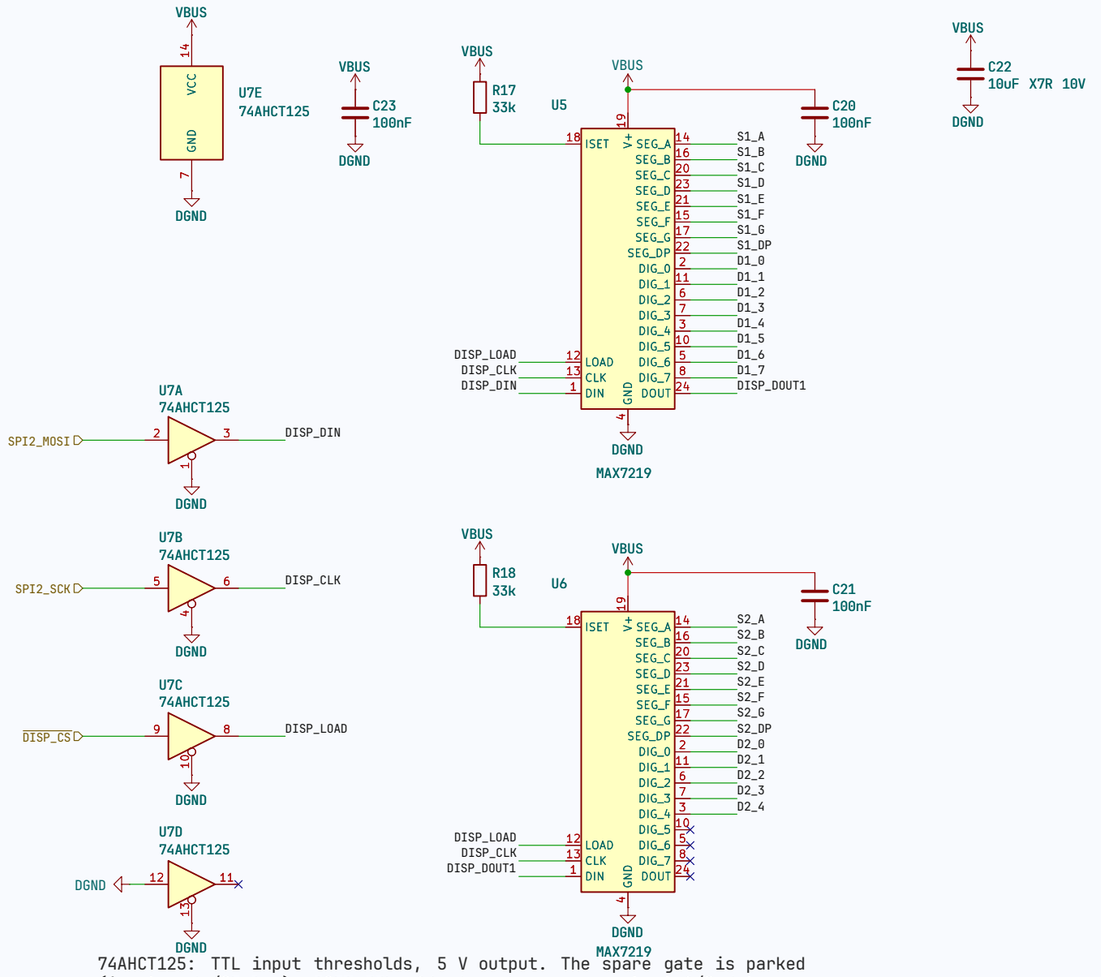
J5 takes a 0.96" SSD1306 module for menus and battery statistics; the 7-segment panel is the primary UI.

DC Load Box
Sheet: /MCU/
File: mcu.kicad_sch
Title: MCU: STM32G0B1CBT6 (LQFP48), USB device, SPI to the isolated stage,
Size: A3 | Date: 2026-10-31 | Rev: D
KiCad E.D.A. 10.0.6 | Id: 3/13

Front panel: 3 x 4-digit 7-segment readouts, 8 status LEDs, encoder, 3 buttons

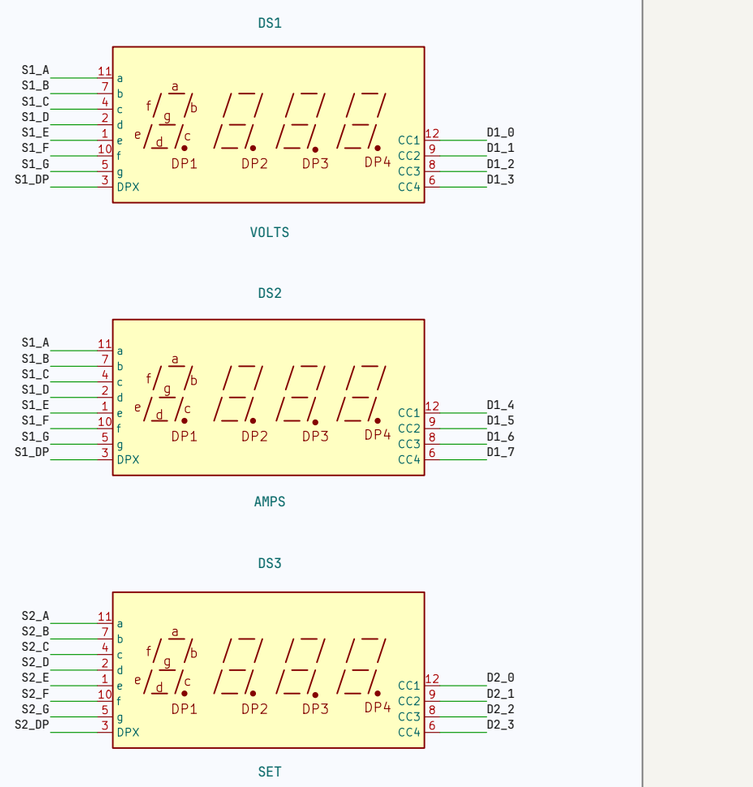
Two MAX7219 in a chain, fed through a 74AHCT125 level shifter (MAX7219 VIH is 3.5 V). Segment and digit nets connect by name.

A Display drivers (5 V) and level shifter



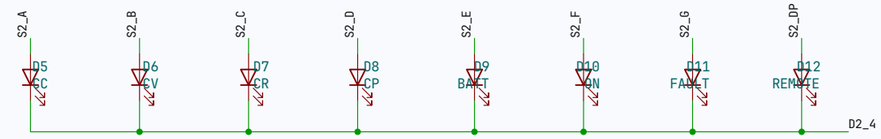
74AHCT125: TTL input thresholds, 5 V output. The spare gate is parked (input and /OE low). ISET 33k: 16 mA peak per segment at 1/8 duty; the intensity register trims it further.

B Readouts: VOLTS, AMPS, SET (0.56" common-cathode, red)



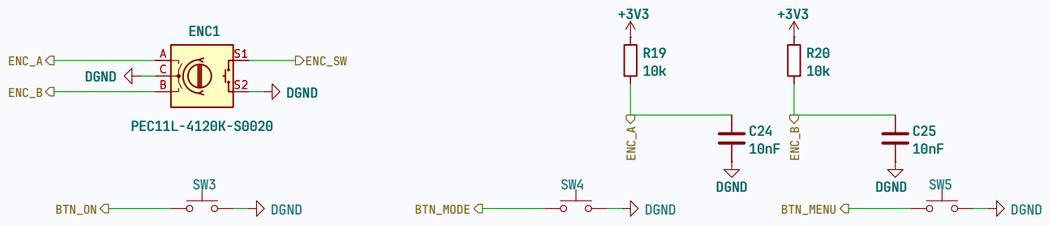
Digit 1 is the left-most. VOLTS 0.000 - 58.00, AMPS 0.000 - 10.00 (0.4 mA per ADC LSB), SET = the value being edited. CC56-12 pin numbers are from the KiCad symbol; check them against the display you buy.

C Status LEDs on U6 digit 4



Anode = segment A..DP, cathode = digit 4 of U6. ON while the load sinks; FAULT for OTP / OVP / NTC-open; REMOTE while a host holds the instrument.

D Encoder and buttons (MCU pull-ups on the buttons, RC on the encoder)



ON/OFF toggles the sink. MODE cycles CC/CV/CR/CP/BATT. MENU: limits, fan, cal. Push = digit.

DC Load Box

Sheet: /Front Panel UI/
File: ui.kicad_sch

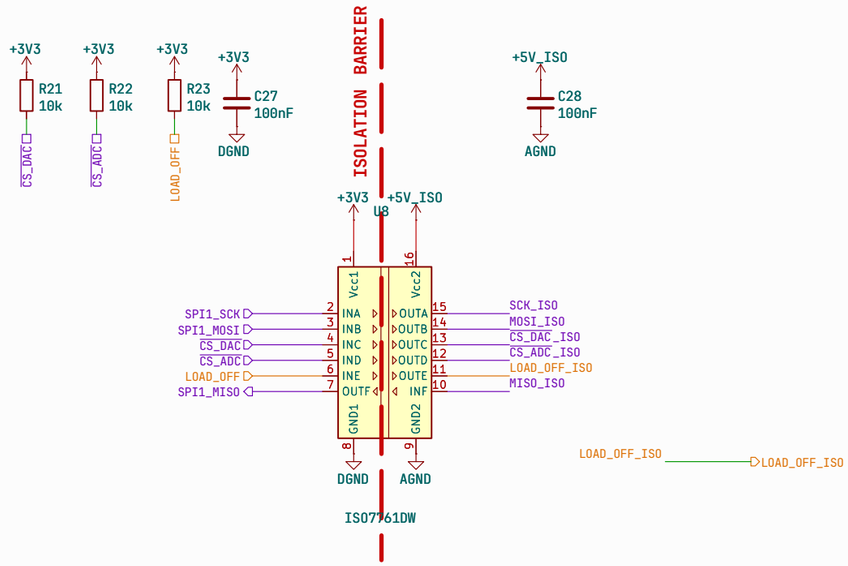
Title: Front panel: 3 x 4-digit 7-segment readouts, 8 status LEDs, encoder,

Size: A3	Date: 2026-10-31	Rev: D
KiCad E.D.A. 10.0.6		Id: 4/13

Isolated interface: digital isolator, set-point DAC, readback ADC

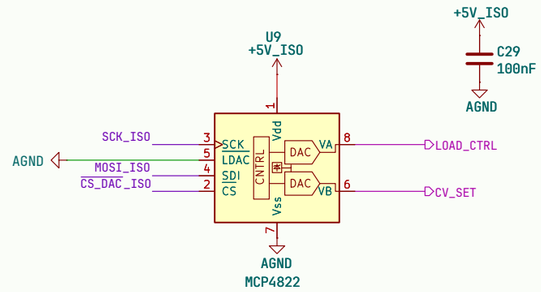
SPI crosses the barrier. The DAC and ADC carry their own references, so no precision reference or trimmer is needed.

A Digital isolator



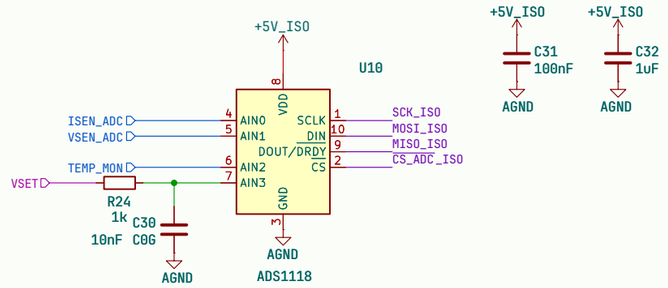
Side 1 = MCU (3.3 V), side 2 = analog (5 V). Five channels go to the analog side, MISO comes back. ISO7761 (no F suffix) outputs default HIGH when side 1 is unpowered or an input floats: chip selects stay de-asserted and LOAD_OFF_ISO is asserted, so the load is OFF until firmware says otherwise. R21-R23 give the same guarantee while the MCU is in reset. Propagation delay 11 ns: SPI at 2 MHz or less.

B Set-point DAC: A = current ceiling, B = CV voltage



12 bit, internal 2.048 V reference, gain 2: 0 - 4.095 V in 1 mV steps. A: LOAD_CTRL, 2.5 A per volt = 2.44 mA per LSB. B: CV_SET in VSEN volts. Both outputs are 0 V after power-up (POR shutdown): no EEPROM surprise like the MCP4725.

C Readback ADC: 16 bit, 4 channels



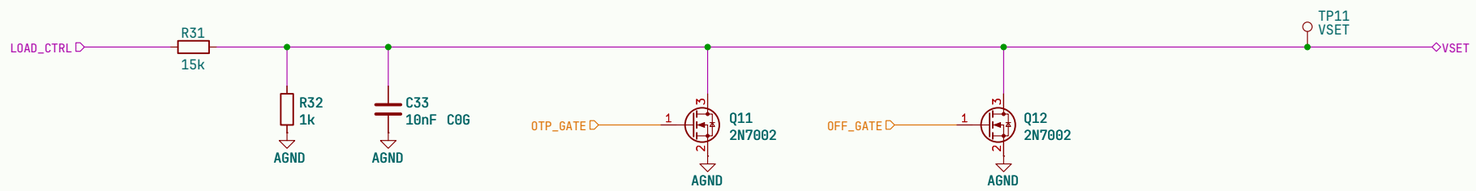
AIN0 ISEN (0.30 V/A), AIN1 VSEN (0.049 V/V), AIN2 heatsink NTC, AIN3 the live V_SET node (PGA +-256 mV): shows what the loops really get after OTP, OFF, KILL or CV pull it down. Internal reference 0.01 %, 7 ppm/K, 860 SPS, plus a die temperature sensor.

DC Load Box
Sheet: /Isolated Interface/
File: iso_interface.kicad_sch
Title: Isolated interface: digital isolator, set-point DAC, readback ADC
Size: A3 | Date: 2026-10-31 | Rev: D
KiCad E.D.A. 10.0.6 | Id: 5/13

Set-point and hardware CV: V_SET divider, protective pull-downs, CV error amplifier

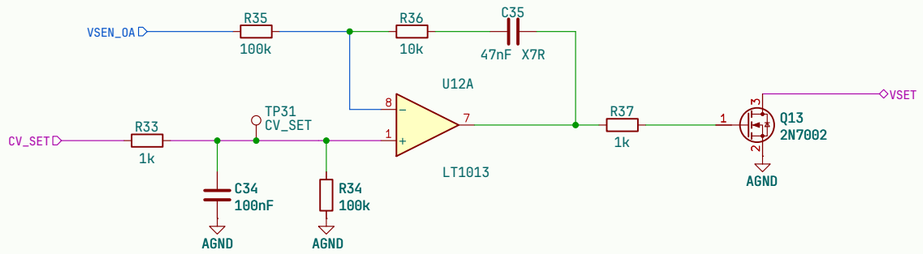
V_SET = LOAD_CTRL / 16 is the only node the four channels follow. Anything that must stop the load pulls it to AGND.

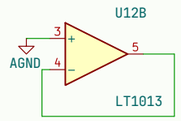
A V_SET: 0 - 256 mV = 0 - 10.24 A



R31 / R32 = 15 : 1, 0.1 %: 4.096 V in gives 256 mV = 2.56 A per channel. C33 slews the step in 9 us so the loops never see a step edge. Q11 and Q12 are hardware and work with the MCU dead. CV (block B) pulls this node down too, so LOAD_CTRL stays the ceiling in CV.

B Hardware constant-voltage loop (LTspice deck 06)





spare half: follower at AGND

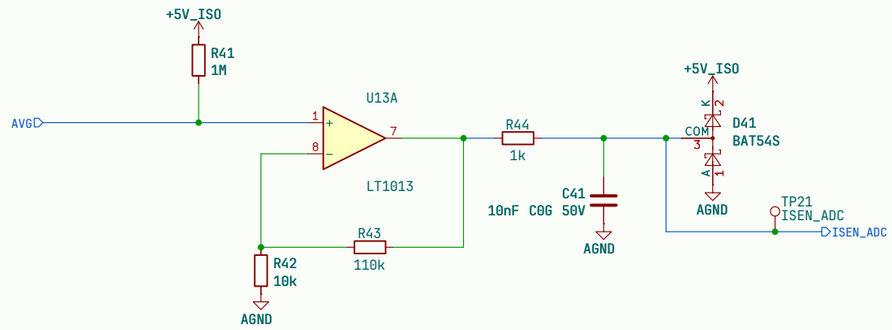
Why: a current-source DUT cannot be held in CC (two current sources in series rail to compliance), and a firmware CV loop on the DAC cannot either: 1 LSB into the 2.2 uF snubber slews 11 V in one 10 ms update. Here: V_DUT below CV_SET → U12A rises → Q13 pulls V_SET down → less current. CV_SET = 0 V: VSEN_OA (85 mV minimum) > CV_SET, U12A stays low, Q13 off, plain CC (decks 01-04). Deck 06: settles for 0.1 - 8 A with and without 47 uF across the DUT. Below 0.1 A it hunts: use CC or a resistor there.

DC Load Box		
Sheet: /Setpoint & CV/		
File: setpoint_cv.kicad_sch		
Title: Set-point and hardware CV: V_SET divider, protective pull-downs, CV e		
Size: A3	Date: 2026-10-31	Rev: D
KiCad E.D.A. 10.0.6		Id: 6/13

Sense: ISEN = mean of the four source nodes, VSEN = true 4-wire difference amplifier

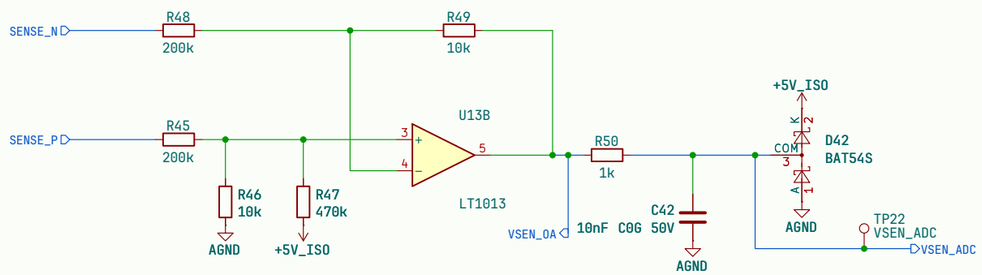
Both outputs are scaled for the 4.096 V ADC range and clamped to +5V_ISO. Offsets are deliberate and calibrated out in firmware.

A ISEN: 0.300 V/A + 0.15 V



Gain 12 (R43 / R42 + 1). R41 injects +12 mV at the input (= +0.15 V out) so the amp never sits on its 35 mV output floor. Full scale 3.22 V at 10.24 A, 0.4 mA per ADS1118 LSB. U13 is the S8 package: NON-STANDARD pinout, the symbol is correct for it.

B VSEN: 0.049 V/V + 0.10 V, true 4-wire



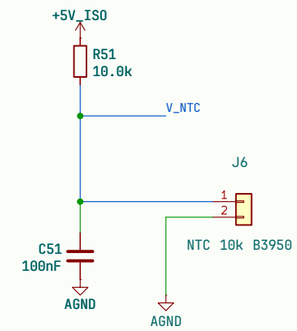
VSEN = 0.049 x (SENSE_P - SENSE_N) + 0.10 V: 2.5 mV of DUT per LSB, clips at 65 V. R45 and R48 see 60 V: 1206. R49 feeds back to -IN (the Rev B board had it on +IN and latched). VSEN_OA also feeds the CV loop and the OVP comparator.

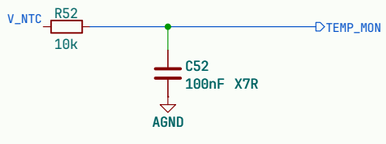
DC Load Box		
Sheet: /Sense/		
File: sense.kicad_sch		
Title: Sense: ISEN = mean of the four source nodes, VSEN = true 4-wire dif		
Size: A3	Date: 2026-10-31	Rev: D
KiCad E.D.A. 10.0.6		Id: 7/13

Protection: over-temperature, dropout / KILL, NTC-open, over-voltage, hardware OFF

LM339 open-collector comparators on +12V_ISO. Thresholds are ratios of +5V_ISO, the same rail that feeds the NTC divider.

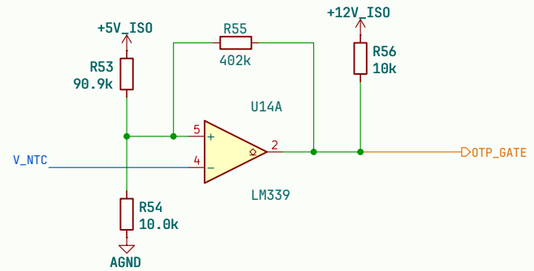
A Heatsink NTC and TEMP_MON





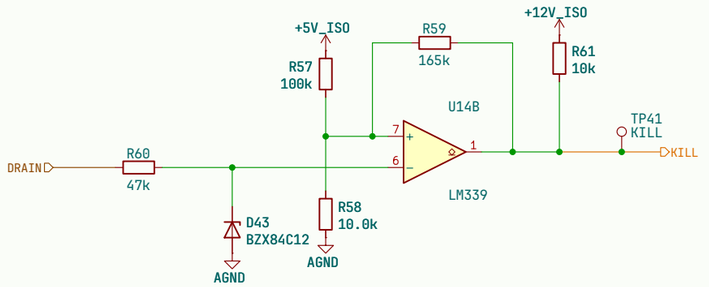
V_NTC (10k B3950 lug thermistor on the heatsink next to FET 2): 2.50 V at 25 C, 1.52 V at 45 C, 0.75 V at 70 C, 0.49 V at 85 C. TEMP_MON goes to the ADC: the MCU runs the fan from it and shows the temperature.

B Over-temperature: 85 C trip, 70 C release



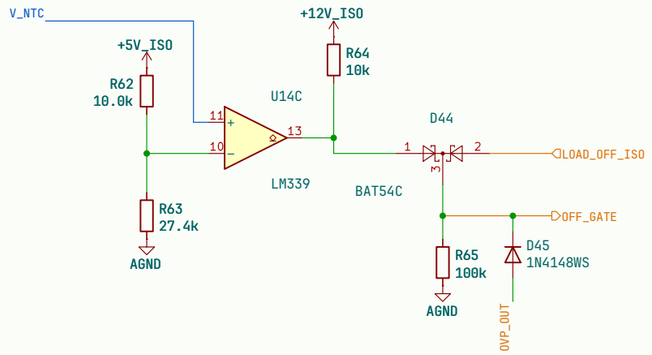
In+ = 0.099 x rail (0.49 V = 85 C). Output high → Q11 shorts V_SET. Hysteresis via R55 from the 12 V output: releases at 0.75 V = 70 C.

C Dropout / anti-wind-up KILL: DRAIN below 0.45 V



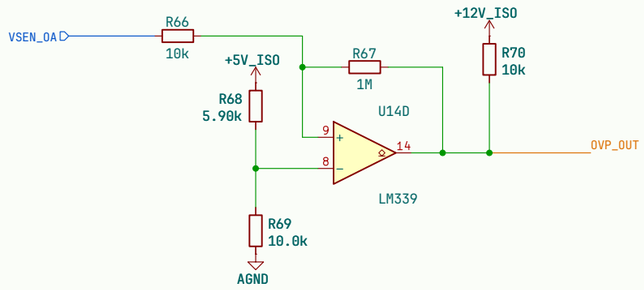
KILL pulls every channel In- high through 4.7k + 1N4148 when the DUT is absent or below 0.45 V; releases above 0.54 V. Stops the 115 - 170 A recovery spikes of deck 04. R60 sees 60 V: 1206.

D NTC open and hardware OFF: all roads to OFF_GATE



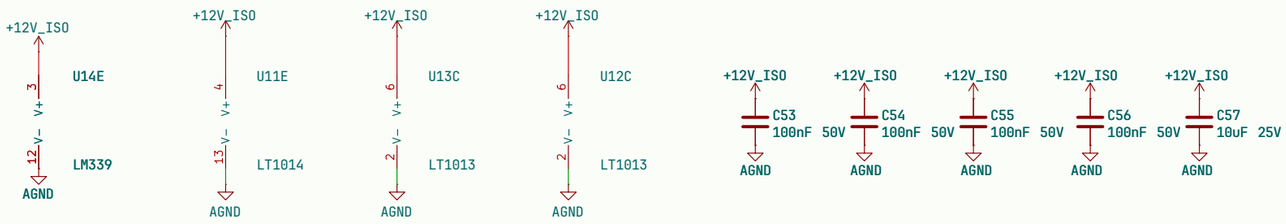
NTC open or unplugged: V_NTC above 0.73 x rail (= 3.66 V, NTC > 27 k) forces 0 A; an open NTC would otherwise read as cold. False trip only below about 3 C. LOAD_OFF_ISO: the MCU-side OFF, 5 V logic through the isolator, asserted whenever the USB side is dead. OVP_OUT (block E) joins through D45. R65 defines the node when nothing drives it.

E Over-voltage: DUT above 62 V



VSEN_OA above 0.629 x rail (3.15 V = 62 V at the terminals) forces 0 A through OFF_GATE. Protects the 58 V fuse rating and the 70 V TVS. R67 gives about 1.8 V of hysteresis. Firmware latches the trip; the comparator alone would re-arm when the DUT sags.

F Supplies of the analog ICs



100 nF 50 V at every V+ pin, 10 uF bulk near the LT1014.

DC Load Box
Sheet: /Protection/
File: protection.kicad_sch

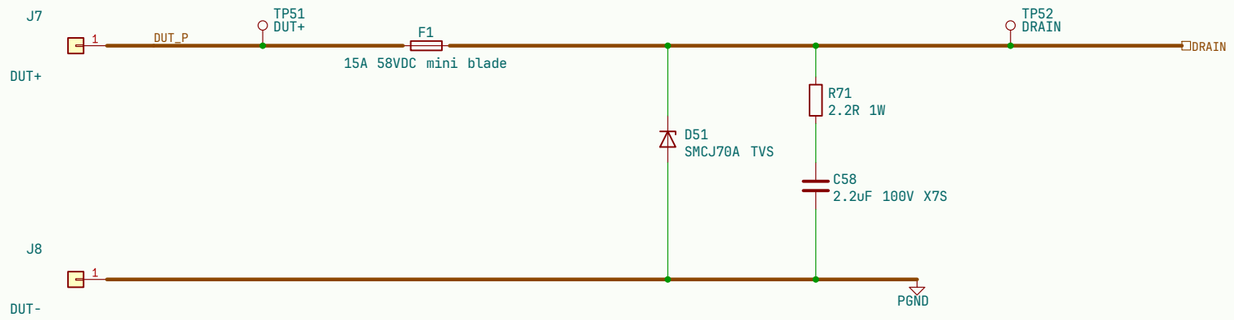
Title: Protection: over-temperature, dropout / KILL, NTC-open, over-voltage,

Size: A3	Date: 2026-10-31	Rev: D
KiCad E.D.A. 10.0.6		Id: 8/13

Power path: DUT terminals, 58 V fuse, TVS, snubber, 4-wire sense

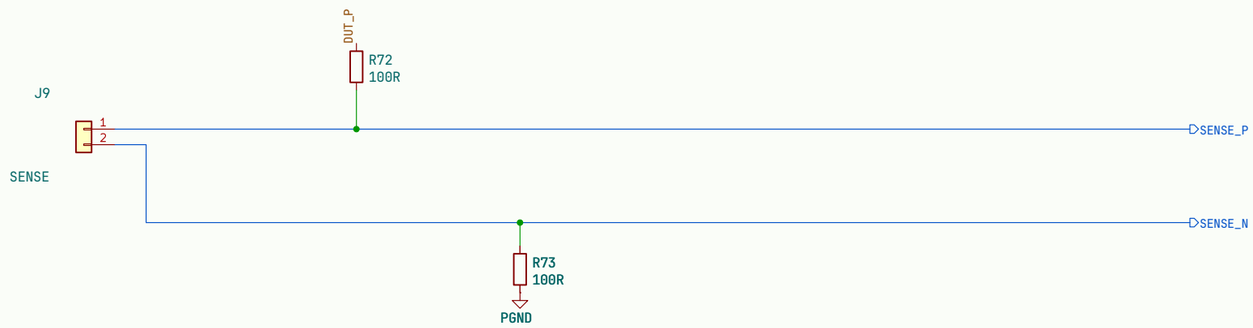
DUT+ → fuse → DRAIN bus → four drains. The source resistors return to PGND, which meets AGND only at NT1.

A Terminals and DRAIN bus



Fuse: the 58 V DC MINI part named. Ordinary blade fuses are rated 32 V, so 58 V is the DUT limit of this board (OVP trips at 62 V). No series diode, so the load works down to 0.6 V. Reverse polarity conducts through the FET body diodes and the TVS and clears the fuse. Copper: ⩾ 5 mm of 2 oz for DUT+ / DRAIN / PGND, or a soldered wire. C58 and D51 directly across DRAIN / PGND at the terminals.

B 4-wire sense with local fallback



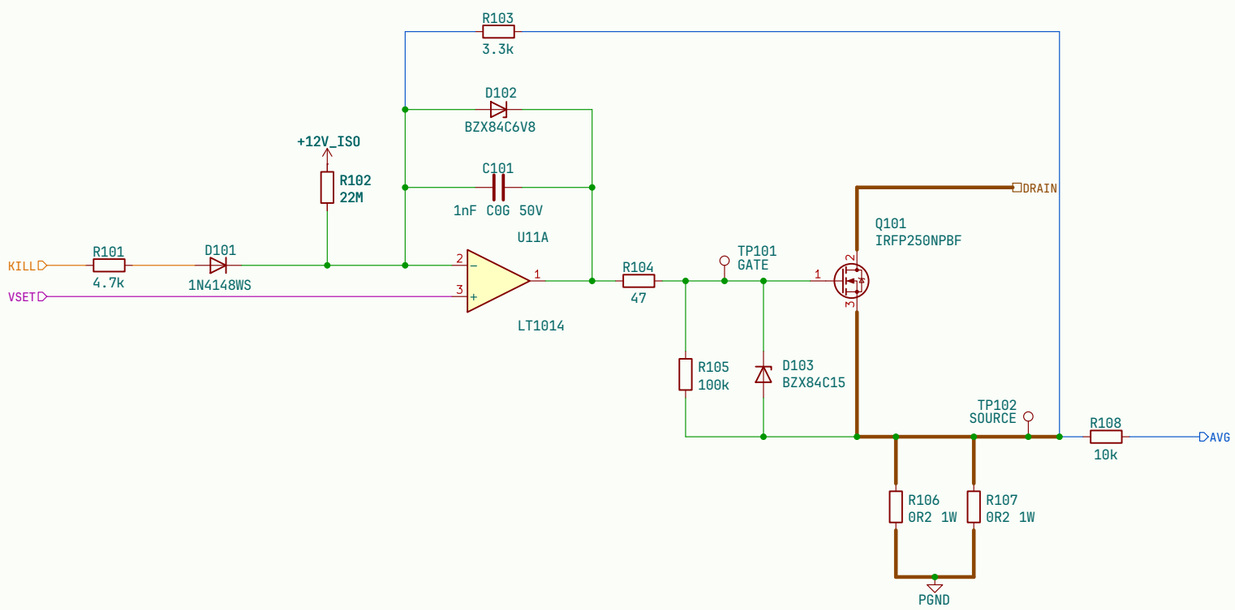
J9 open: R72 / R73 connect the sense amplifier to the terminals. J9 wired to the DUT: they carry only microamps and the lead drop leaves the reading. A miswired sense lead burns a 100 R 1206 open and nothing else.

DC Load Box		
Sheet: /Power Path/		
File: power_path.kicad_sch		
Title: Power path: DUT terminals, 58 V fuse, TVS, snubber, 4–wire sense		
Size: A3	Date: 2026–10–31	Rev: D
KiCad E.D.A. 10.0.6		Id: 9/13

Current-sink channel (4 instances, one LT1014 section each)

Each MOSFET gets its own loop forced to V_SET / 0.1 R, so current and power share equally whatever the threshold spread.

Channel (one of four): LT1014 section, 3.3k / 1n / 47R compensation, IRFP250N, 2 x 0.2R Kelvin sense



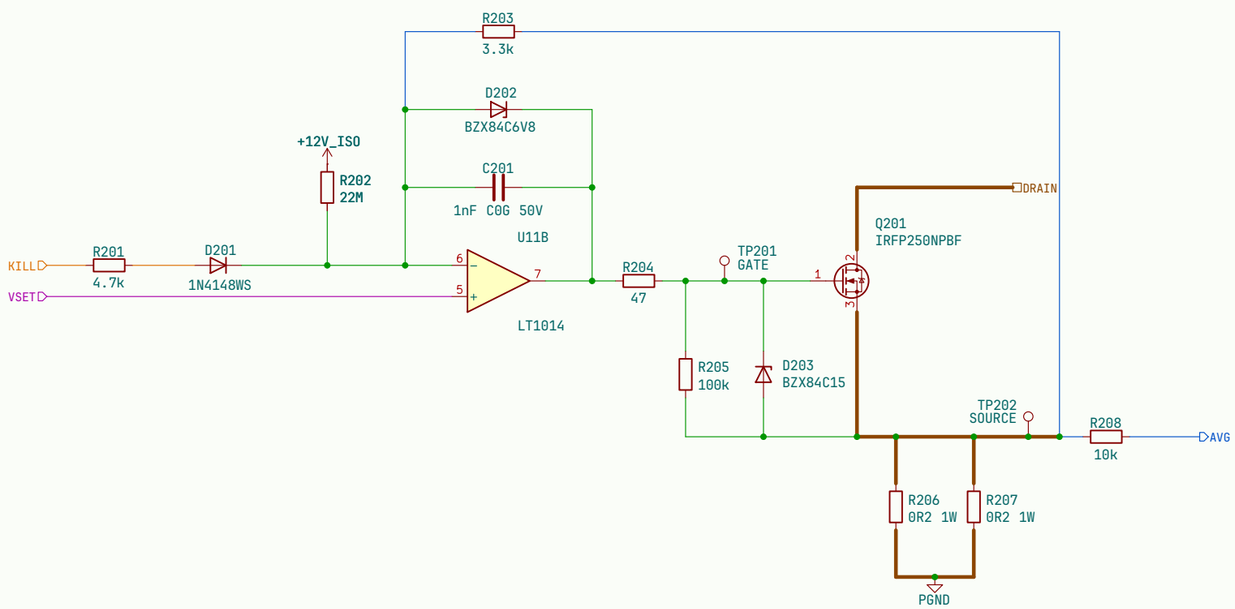
Rs: two 0.2 R 2512 in parallel = 0.1 R, so I_ch = V_SET / 0.1 R (2.56 A full scale).
KELVIN: Rn03 and Rn08 connect at the TOP PAD of the Rs pair on their own trace, never along the high-current source copper. Rg within 10 mm of the gate pin. Dn03 and Rn05 sit at the FET pins. Dn02 limits integrator wind-up; Rn02 gives +1.8 mV offset so DAC code 0 is truly 0 A (dead zone about 72 mA in total). Loop values from deck 03: PM ≥ 80 deg, GM ≥ 13 dB for every FET candidate. Change them only with a re-run of deck 03.
(n = channel number)

DC Load Box		
Sheet: /Channel 1/		
File: channel.kicad_sch		
Title: Current-sink channel (4 instances, one LT1014 section each)		
Size: A3	Date: 2026-10-31	Rev: D
KiCad E.D.A. 10.0.6		Id: 10/13

Current-sink channel (4 instances, one LT1014 section each)

Each MOSFET gets its own loop forced to V_SET / 0.1 R, so current and power share equally whatever the threshold spread.

Channel (one of four): LT1014 section, 3.3k / 1n / 47R compensation, IRFP250N, 2 x 0.2R Kelvin sense



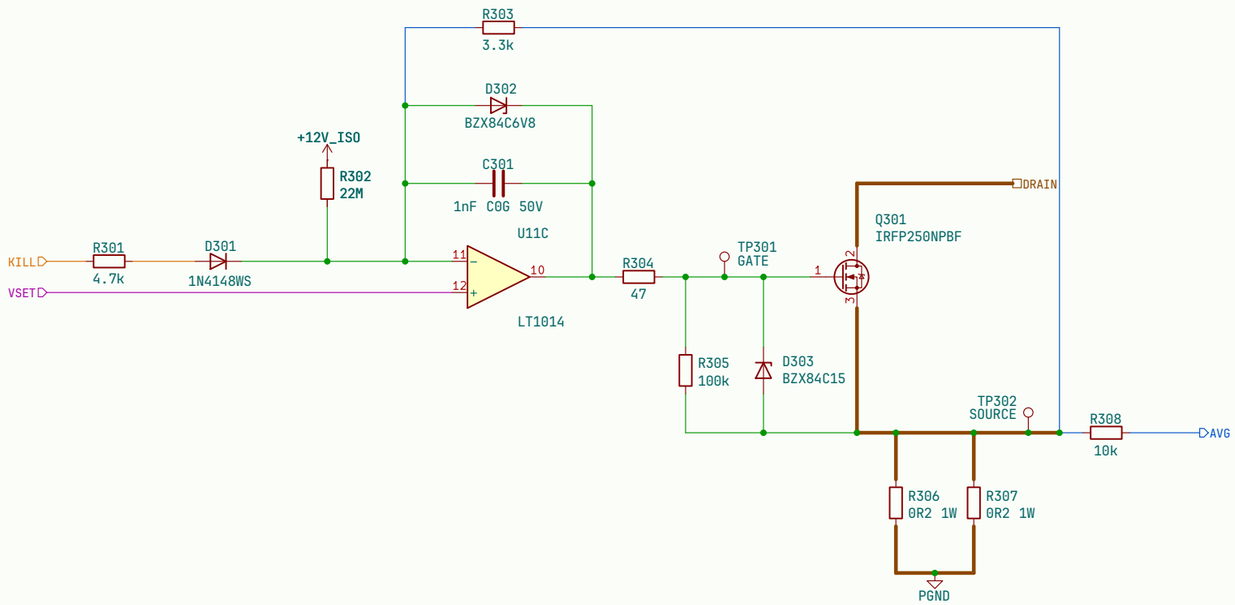
Rs: two 0.2 R 2512 in parallel = 0.1 R, so I_ch = V_SET / 0.1 R (2.56 A full scale).
KELVIN: Rn03 and Rn08 connect at the TOP PAD of the Rs pair on their own trace, never along the high-current source copper. Rg within 10 mm of the gate pin. Dn03 and Rn05 sit at the FET pins. Dn02 limits integrator wind-up; Rn02 gives +1.8 mV offset so DAC code 0 is truly 0 A (dead zone about 72 mA in total). Loop values from deck 03: PM ≥ 80 deg, GM ≥ 13 dB for every FET candidate. Change them only with a re-run of deck 03.
(n = channel number)

DC Load Box		
Sheet: /Channel 2/		
File: channel.kicad_sch		
Title: Current-sink channel (4 instances, one LT1014 section each)		
Size: A3	Date: 2026-10-31	Rev: D
KiCad E.D.A. 10.0.6		Id: 11/13

Current-sink channel (4 instances, one LT1014 section each)

Each MOSFET gets its own loop forced to V_SET / 0.1 R, so current and power share equally whatever the threshold spread.

Channel (one of four): LT1014 section, 3.3k / 1n / 47R compensation, IRFP250N, 2 x 0.2R Kelvin sense



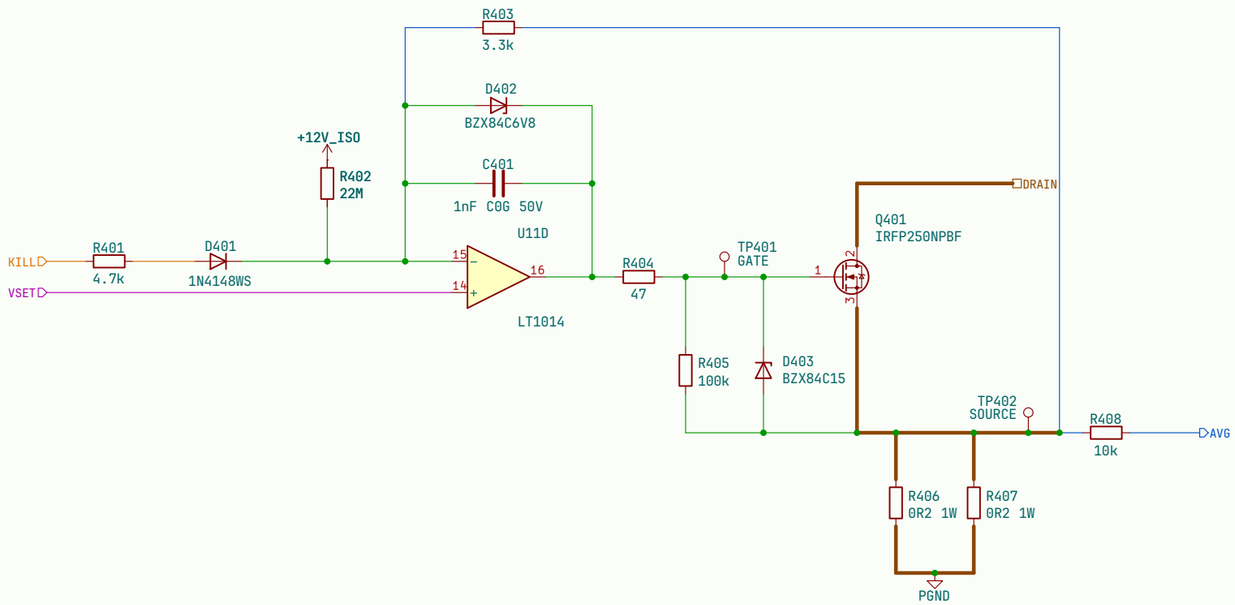
Rs: two 0.2 R 2512 in parallel = 0.1 R, so I_ch = V_SET / 0.1 R (2.56 A full scale).
KELVIN: Rn03 and Rn08 connect at the TOP PAD of the Rs pair on their own trace, never along the high-current source copper. Rg within 10 mm of the gate pin. Dn03 and Rn05 sit at the FET pins. Dn02 limits integrator wind-up; Rn02 gives +1.8 mV offset so DAC code 0 is truly 0 A (dead zone about 72 mA in total). Loop values from deck 03: PM ≥ 80 deg, GM ≥ 13 dB for every FET candidate. Change them only with a re-run of deck 03.
(n = channel number)

DC Load Box		
Sheet: /Channel 3/		
File: channel.kicad_sch		
Title: Current-sink channel (4 instances, one LT1014 section each)		
Size: A3	Date: 2026-10-31	Rev: D
KiCad E.D.A. 10.0.6		Id: 12/13

Current-sink channel (4 instances, one LT1014 section each)

Each MOSFET gets its own loop forced to V_SET / 0.1 R, so current and power share equally whatever the threshold spread.

Channel (one of four): LT1014 section, 3.3k / 1n / 47R compensation, IRFP250N, 2 x 0.2R Kelvin sense



Rs: two 0.2 R 2512 in parallel = 0.1 R, so I_ch = V_SET / 0.1 R (2.56 A full scale).
KELVIN: Rn03 and Rn08 connect at the TOP PAD of the Rs pair on their own trace, never along the high-current source copper. Rg within 10 mm of the gate pin. Dn03 and Rn05 sit at the FET pins. Dn02 limits integrator wind-up; Rn02 gives +1.8 mV offset so DAC code 0 is truly 0 A (dead zone about 72 mA in total). Loop values from deck 03: PM ≥ 80 deg, GM ≥ 13 dB for every FET candidate. Change them only with a re-run of deck 03.
(n = channel number)

DC Load Box		
Sheet: /Channel 4/		
File: channel.kicad_sch		
Title: Current-sink channel (4 instances, one LT1014 section each)		
Size: A3	Date: 2026-10-31	Rev: D
KiCad E.D.A. 10.0.6		Id: 13/13